

Unit IV

IoT Platform and its Programming

4

4.1 : Introduction to Arduino

Q.1 What is an Arduino ? Explain its feature.

Ans. : • Arduino is an open-source electronics platform based on easy-to-use hardware and software.

- Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online.
- The Arduino Uno can be powered via the USB connection or with an external power supply. The power source is selected automatically.

Features :

- Support fast computations, ARM based MCU
- AVR micro-controller clock is ATSAM3X81
- Operating input voltages is 3.3 Volt
- It uses EEPROM, SRAM and Flash memory
- It also support USB and UART

Q.2 Briefly discuss pin of Arduino.

Ans. : • The pins on the Arduino can be configured as either inputs or outputs. Uno R3 includes 14-digital pins which can be used as an input otherwise output by using the functions like pin Mode (), digital Read(), and digital Write().

- These pins can operate with 5V, and every digital pin can give or receive 20mA, and includes a 20k to 50k ohm pull up resistor. The maximum current on any pin is 40mA which cannot surpass for

Internet of Things 4-2 IoT Platform and its Programming

Arduino Pin Description

LED	It comes with built-in LED connected through pin 13. Providing HIGH/ LOW value to the pin will turn it ON / OFF.
Vin	It is a 5V input / supply voltage pin to the Arduino Board through a USB port. USB (5V), Vin (7V to 20V) pin, or DC power jack are the 3 ways to provide the power for the board. More voltage provided damages the board.
GND	Ground pins.
Reset	Used to the program running on the board.
PWM	Pins (3,6,9,10, 11) is provided by PWM which provides 8-bit output PWM. SPI It is known as Serial Peripheral Interface. Four pins (10(SS), 11(MOSI), 12(MISO), 13(SCK)) provide SPI communication with the help of SPI library.
AREF	Analog Reference (AREF) pin is used for providing a reference voltage to the analog inputs.
TWI	A4, A5 pins are used for Two-Wire Interface (TWI). TWI communication is accessed through wire library.
Rx	It is used to receive data. i.e. used of serial communication.
TX	It is used to transmit data. i.e. used of serial communication.
External Interrupts	Pins 2 and 3 are used for providing external interrupts by providing LOW or changing value.

4.2 : Raspberry Pi

Q.3 What is Raspberry Pi ?

Ans. : Raspberry Pi is the name of a series of single-board computers made by the Raspberry Pi Foundation. The original Pi had a single core

700 MHz CPU and just 256 MB RAM, and the latest model has a quad-core 1.4 GHz CPU with 1 GB RAM.

Q.4 What are the different Raspberry Pi model types ?

Ans. : The Raspberry Pi models are of two types :

1. Model A (Introduced later as a hardware-reduced model)
2. Model B (Introduced first and is the full hardware model)

Q.5 Describe features of Raspberry Pi board.

Ans. : • Features of Raspberry Pi board (Model B - 2 Version) :

1. This second generation Raspberry Pi has an upgraded Broadcom BCM2836 processor, which is a powerful ARM Cortex-A7 based quad-core processor that runs at 900 MHz. The board also features an increase in memory capacity to 1 GB.

2. Ethernet port
3. Two USB ports
4. Two video output options : HDMI or composite
5. 3.5 mm audio output jack
6. 26-pin GPIO header with 0.1" spaced male pins that are compatible with our 2x13 stackable female headers and the female ends of our premium jumper wires.
7. Display Serial Interface (DSI) 15 way flat flex cable connector with two data lanes and a clock lane.

Q.6 How is Raspberry Pi different from a desktop computer ?

Ans. : • In Raspberry Pi, operating system is installed on SD card whereas in desktop computer, operating system is installed in hard disk.

- Raspberry Pi does not have their own CPU and RAM.
- Processing power of Raspberry Pi is less as compared to desktop computers.
- Raspberry Pi uses less power than desktop computers.

Q.7 Explain difference between Model A and Model B of Raspberry Pi.

Ans. : Difference between Model A and Model B of Raspberry Pi :

Parameters	Model A	Model B
GPU type	VideoCore IV	VideoCore IV
USB port	1	2
Memory	256 MB	512 MB
Ethernet port	No Ethernet port	10/100 Ethernet
SoC Type	Broadcom BCM2837B0	Broadcom BCM2837B0
Number Of Cores	4	4
Type	It is hardware-reduced model	It is full hardware model

Q.8 What is Raspberry Pi ? Explain about its versions and various interfaces in detail.

Ans. : • A Raspberry Pi is a credit card-sized computer originally designed for education, inspired by the 1981, BBC Micro.

- Creator Eben Upton's goal was to create a low-cost device that would improve programming skills and hardware understanding at the pre-university level.
- The Raspberry Pi is slower than a modern laptop or desktop but is still a complete Linux computer and can provide all the expected abilities that implies, at a low-power consumption level.

Versions	Remarks
Raspberry Pi 1	• The original Raspberry Pi had 256 MB of RAM, which increased to 512 MB in a later revision. • It has a 26-way GPIO connector.
Pi Zero	• The Pi Zero includes the GPIO connector, but the header pins are not soldered.
Raspberry Pi 2	• The Raspberry Pi 2 swapped the single-core processor for a much faster quad-core processor and increased the memory to 1 GB RAM

Raspberry

Raspberry Pi 3

- The Raspberry Pi 3 changes the processor to an even more powerful 64-bit processor.
- It also adds Wi-Fi and bluetooth which previously needed to be added as a USB device.
- The Raspberry Pi 3 Model B was launched in February 2016.

• To get the Raspberry Pi working an SD card needs to be prepared with the Linux operating system installed.

• Raspberry Pi users have made many creative and impressive projects using this device. It can also be programmed to assist in 'housekeeping' your network by functioning as NAS, LDAP server, web server, media server, DNS server etc.

• The Raspberry Pi Foundation recommends Python. Any language which will compile for ARMv6 can be used. It installed by default on the Raspberry Pi: C, C++, Java, Scratch and Ruby.

Q.9 Draw and explain Raspberry Pi block diagram and its components.

Ans: [SFPU : May-19, End Sem, Marks 8]

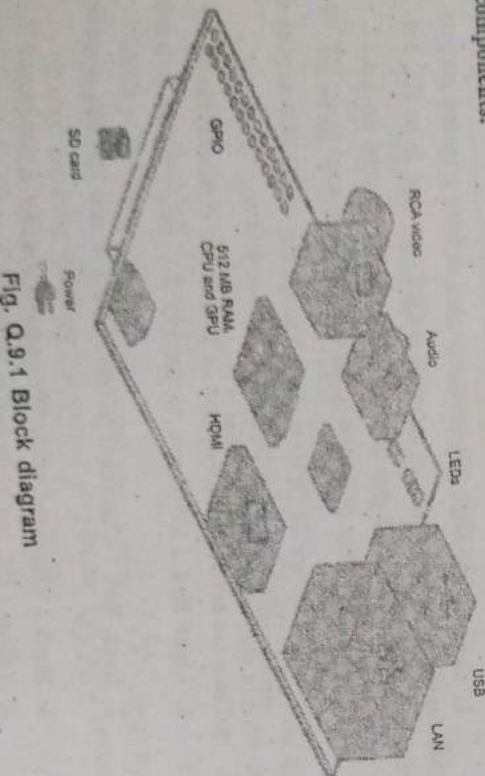


Fig. Q.9.1 Block diagram

Ans. :

Fig. Q.9.1 shows the Raspberry Pi board.

• The Raspberry Pi does not have a separate CPU, RAM or GPU. Instead they are all squeezed into one component called a system on chip or SoC unit.

• Raspberry Pi is open hardware with the exception of its primary chip, the Broadcom SoC which runs the main components of the board - CPU, graphics, memory, USB controller etc.

• All of these Raspberry Pi models share the following features :

1. Operating systems : Raspbian RaspBMC, Arch Linux, Rise OS, Open ELEC Pidora
2. Video output : HDMI Composite RCA
3. Supported resolutions : 640x350 to 1920x1200, including 1080p, PAL and NTSC standards
4. Power source : Micro USB

Components	Description
Processor	• Raspberry Pi uses an ARM processor which is also installed in a wide variety of mobile phones. • This CPU is single core, however it does have a co-processor to perform floating point calculations.
Memory	• Model B Raspberry Pi has 512 MB SDRAM (Synchronous Dynamic RAM). • It store programs that are currently being run in the CPU.
USB ports	• Board has two USB ports. USB port can provide a current upto 100 mA. • Using powered hub, it is possible to connect more devices.
HDMI Output	• High Definition Multimedia Interface (HDMI) supports high-quality digital video and audio through a single cable. • It is also possible to connect a computer monitor with a DVI connection to HDMI using a converter.
Composite Video Output	• It support composite video output with RCA jack and also support PAL and NTSC.

	The TV/DAC pin can be used to output composite video
Audio Output	- Audio output jack is 3.5 mm. - This jack is used for providing audio output to old television along with the RCA jack for video
GPIO Pins	- Both models have a total of 26 GPIO pins, organized into one pin header, named the P1 header - The newer Raspberry Pi adds 8 more GPIO pins in a new pin header called P5 - Not all the GPIO pins are programmable. Some of them are 5.0 VDC or 3.3 VDC positive power pins, some of them are negative ground pins and a few of them are marked DNC (do not connect). - The P1 header has 17 programmable pins and the P5 header adds 4 more. - Reading from various environmental sensors. Writing output to dc motors, LEDs for status.
Power Input	Micro-USB connector is used for power input.
Status LED	It has five status LED.
CSI	Camera Serial Interface (CSI) can be used to connect a camera module to Raspberry Pi.
SD Card Slot	This card is used for loading operating system.

Q.10 What is the use of GPIO pins ?

Ans. : • The Raspberry Pi comes with a set of 26 exposed vertical pins on the board. These pins are a General Purpose Input/Output (GPIO) interface that is purposely not linked to any specific native function on the Raspberry Pi board.

• Instead, the GPIO pins are there explicitly for the end user to have low-level hardware access directly to the board for the purposes of attaching other hardware boards, peripherals, LCD display screens and other hardware devices to the Pi.

ORCODE

• The Raspberry Pi draws its power from a micro-USB port and requires a micro USB to AC adapter. Because the Pi is a micro computer and not simply a cell phone getting a battery topped off, you need to use a high quality charger with stable power delivery that provides a consistent 5 V with at least 700 mA minimum output for older model units and 2.5 A for the Pi 3.

4.3 : Raspberry Pi Interface

Q.11 Write short note on GPIO of Raspberry Pi.

Ans. : • GPIO stands for General Purpose Input/Output (GPIO) can be an input, an output or operate as a serial port Tx data line. The GPIO pins allow you to interface the Pi to the real world.

• Raspberry Pi is a 40-pin expansion header. The pins are numbered from 1 to 40, with the even numbers on the outside edge of the board as shown below.

• There are 4 different types of GPIO pins on the Raspberry Pi. Of these, three may be used as GPIO or special purposes.

1. Green : These are generic GPIO pins without a special purpose.
2. Blue : These pins may operate as GPIO or used as an I2C bus.
3. Orange : These pins may operate as GPIO or used for UART.
4. Purple : These pins may operate as GPIO or used as an SPI bus.

• In total there are 24 GPIO ports. Although any GPIO pins may be used, when possible the green pins should be used first.

• There are several different types of pins in the expansion header :

1. Light Red : Connected to the Pi's 5 V rail, provides a consistent + 5.0 V.
2. Dark Red : Connected to the Pi's 3.3 V rail, provides a consistent + 3.3 V.
3. Black : Connected to the Pi's ground, provides a consistent 0 V.
4. Yellow : I2C bus dedicated to expansion boards

ORCODE

Pin#	NAME		NAME	Pin#
01	3.3V DC Power		DC Power 5V	02
03	GPIO02 (SDA1, I2C)		DC Power 5V	04
05	GPIO03 (SCL1, I2C)		Ground	06
07	GPIO04 (GPIO_GCLK)		(TXD0) GPIO14	08
09	Ground		(RXD0) GPIO15	10
11	GPIO17 (GPIO_GEN0)		(GPIO_GEN1) GPIO18	12
13	GPIO27 (GPIO_GEN2)		Ground	14
15	GPIO22 (GPIO_GEN3)		(GPIO_GEN4) GPIO23	16
17	3.3V DC Power		(GPIO_GEN5) GPIO24	18
19	GPIO10 (SPI_MOSI)		Ground	20
21	GPIO09 (SPI_MISO)		(GPIO_GEN6) GPIO25	22
23	GPIO11 (SPI_CLK)		(SPI_CE0_N) GPIO8	24
25	Ground		(SPI_CE1_N) GPIO07	26
27	ID_SD (I2C ID EEPROM)		(I2C ID EEPROM) ID_SC	28
29	GPIO05		Ground	30
31	GPIO06		GPIO12	32
33	GPIO13		Ground	34
35	GPIO19		GPIO16	36
37	GPIO26		GPIO20	38
39	Ground		GPIO21	40

Fig. Q.11.1 GPIO pin diagram

- All the remaining pins are known as the General Purpose Input/Output (GPIO). The GPIO is a generic pin on a circuit whose behaviour, including whether it is an input or output, may be controlled by the user at run-time.
- The GPIO pins are connected to the +3.3 V rails. When used for input they are able to read voltages. When used for output they may be set to +3.3 V (high) or 0 V (low).

GPIO in Python :

- The easiest way to control the GPIO pins is using the RPi.GPIO Python library. Installing the library is easy if you follow RPi.GPIO installation guide. Once installed using the pins is as easy as :
import RPi.GPIO as GPIO
Use GPIO numbers not pin numbers
GPIO.setmode(GPIO.BCM)

set up the GPIO channels - one input and one output
GPIO.setup(7, GPIO.IN)
GPIO.setup(8, GPIO.OUT)

input from GPIO7
input_value = GPIO.input(7)

output to GPIO8
GPIO.output(8, True)

• In this example we use GPIO7 (pin 26) and GPIO8 (pin 24).

Q.12 What is GND in GPIO ?

Ans. : • GND means ground pins. Ground GPIO pins are physical numbers 6, 9, 14, 20, 25, 30, 34 and 39.

Q.13 Define Raspberry Pi hardware.

Ans. : Raspberry Pi hardware includes ARM processor, GPU, RAM and USB port.

Q.14 What is use of SPI and I2C interfaces on Raspberry Pi ?

Ans. : • Three types of interface is supported by Raspberry Pi.

1. Serial

- It uses serial peripherals for serial communication.

2. Serial Peripheral Interface (SPI)

- SPI is a communication protocol used to transfer data between micro-computers like the Raspberry Pi and peripheral devices. These peripheral devices may be either sensors or actuators.

- SPI uses 4 separate connections to communicate with the target device. These connections are the serial Clock (CLK), Master Input Slave Output (MISO), Master Output Slave Input (MOSI) and Chip Select (CS).

- The clock pin sense pulses at a regular frequency, the speed at which the Raspberry Pi and SPI device agree to transfer data to each other.
- For the ADC, clock pulses are sampled on their rising edge, on the transition from low to high.
- The MISO pin is a data pin used for the master to receive data from the ADC. Data is read from the bus after every clock pulse.
- The MOSI pin sends data from the Raspberry Pi to the ADC. The ADC will take the value of the bus on the rising edge of the clock. This means the value must be set before the clock is pulsed.
- The chip select line chooses which particular SPI device is in use. If there are multiple SPI devices, they can all share the same CLK, MOSI and MISO.

3. I2C

- I2C is a communication protocol that the Raspberry Pi can use to speak to other embedded devices (temperature sensors, displays, accelerometers, etc).
- I2C is a useful bus that allows data exchange between microcontrollers and peripherals with a minimum of wiring.
- I2C is a two wire bus, the connections are called SDA (Serial Data) and SCL (Serial Clock). Each I2C bus has one or more masters (Raspberry Pi) and one or more slave devices, like the I/O Expander.
- As the same data and clock lines are shared between multiple slaves, we need some way to choose which device to communicate with.

Q.15 Describe features of SPI.

Ans. : • The SPI has the following features :

1. 16-bit shift register
2. 16-bit Receive buffer register (SPIBUF) and 16-bit Receive buffer emulation alias register (SPIEMU)
3. 16-bit Transmit data register (SPIDAT0) and 16-bit Transmit data and format selection register (SPIDAT1)
4. 8-bit baud clock generator

5. Serial clock (SPICLK) I/O pin
6. Slave in, master out (SPISIMO) I/O pin
7. Slave out, master in (SPISOMI) I/O pin
8. Multiple slave chip select (SPISCS[n]) I/O pins (4 pin mode only)
9. Programmable SPI clock frequency range
10. Programmable character length (2 to 16 bits)
11. Programmable clock phase (delay or no delay)
12. Programmable clock polarity (high or low)
13. Interrupt capability
14. DMA support (read/write synchronization events)
15. Up to 66 MHz operation

4.4 : Introduction to Python Program with Raspberry Pi

Q.16 How Raspberry Pi controls LED ?

Ans. : Fig. Q.16.1 shows diagram of connecting LED to Raspberry Pi. The LED will initially be off because the GPIO pins are initialized as inputs at power-on.

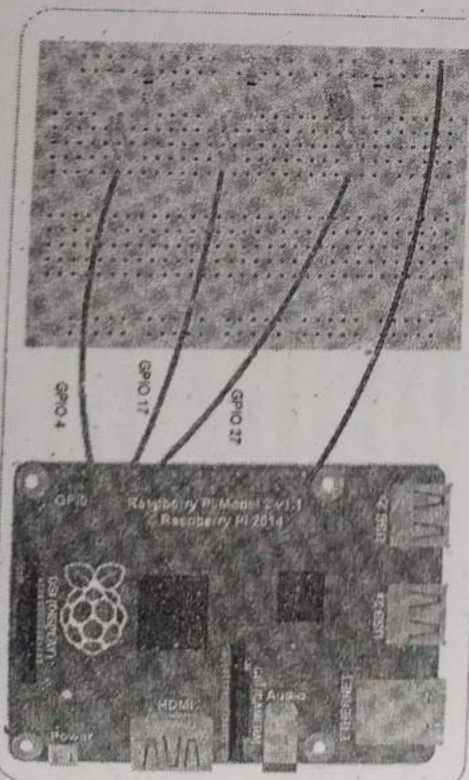


Fig. Q.16.1 Diagram of connecting LED to raspberry pi

- Install Python 2 library Rpi.GPIO. A library that will let us control the GPIO pins. Install commands:

```
sudo apt-get update
```

```
sudo apt-get install python-dev
```

```
sudo apt-get install python-rpi-gpio
```

- Simple LED circuit is shown below :

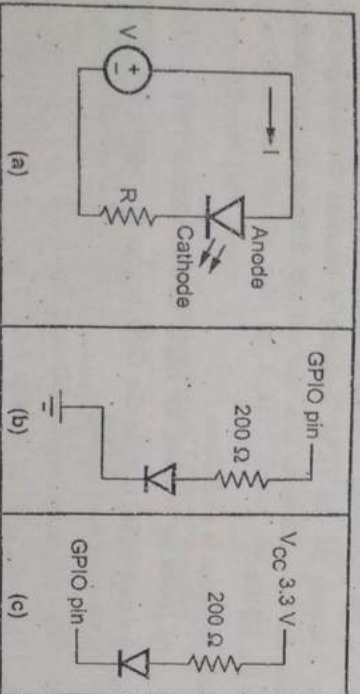


Fig. Q.16.2

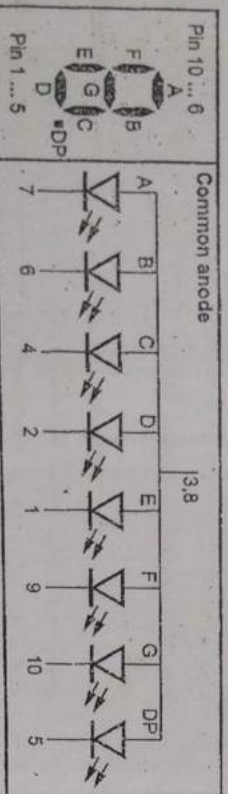


Fig. Q.16.3

- Current flows from the anode (+) to cathode (-). Anode is longer pin and cathode is shorter pin.
- Open up IDLE, the Python programming software and create a New file. Save it as led.py and input the code from the code listing. What the code does is first tell Python to use the GPIO module so we can connect to the GPIO pins, by importing the module.

- We then import the time module so we can create a delay between commands. We then tell the code to treat the GPIO pins as the number they are on the board and to turn the seventh pin into an output.

- We alternate between True and False so that it turns the pin on and off. Once it's cycled a few times, it will print the message 'Done' into IDLE and finally turn off the GPIO pins.

Import RPi.GPIO as GPIO

Import time

GPIO.setmode(GPIO.BOARD)

GPIO.setup(7, GPIO.OUT)

GPIO.output(7, True)

time.sleep(1)

GPIO.output(7, False)

time.sleep(1)

GPIO.output(7, True)

time.sleep(1)

GPIO.output(7, False)

print "Done"

GPIO.cleanup()

Task 1 :

Turn LED on for 2 seconds and off for 1 second, loop forever. Code is given below :

In this example, we use diagram (b), i.e. controlling the LED by controlling the voltage at the anode (+).

```
import RPi.GPIO as GPIO

import time

def main():
    GPIO.cleanup()

    GPIO.setmode(GPIO.BOARD)

    # to use Raspberry Pi board pin numbers
    GPIO.setup(11, GPIO.OUT)

    # set up GPIO output channel
```



```
while True :
    GPIO.output(11, GPIO.LOW)
```

```
# set RPi board pin 11 low. Turn off LED.
```

```
time.sleep(1)
```

```
GPIO.output(11, GPIO.HIGH)
```

```
# set RPi board pin 11 high. Turn on LED.
```

```
time.sleep(2)
```

```
main()
```

• Example : Display digit on 7-segment LED. It is most direct way to control display :

1. Connect pin 3/8 of 7-seg-LED to Vcc
2. Connect the other 8 pins to 8 GPIO pins
3. Configure the 8 GPIO pins as out

• For example : display "2". Turn on segments A, B, D, E, G and turn off segments C, F, DP. Set A, B, D, E, G to LOW and set C, F, DP to HIGH. Set Pin 7, 6, 2, 1, 10 LOW and Set pin 4, 9, 5 HIGH

Q.17 What is Raspbian OS ?

Ans. : • Raspbian is a free operating system based on Debian optimized for the Raspberry Pi hardware.

• An operating system is the set of basic programs and utilities that make your Raspberry Pi run.

• However, Raspbian provides more than a pure OS, it comes with over 35,000 packages, pre-compiled software bundled in a nice format for easy installation on your Raspberry Pi. As Raspbian is a Linux operating system it has good security features, has excellent networking capabilities.

• Raspbian is highly optimized for the Raspberry Pi line's low-performance ARM CPUs.

Q.18 Explain boot sequence of Raspberry Pi.

Ans. : • When you power on your Raspberry Pi, the first bits of code to run is stored in a ROM chip in the SoC and is built into the Pi during manufacturing. This is called the first-stage boot-loader.

• The first-stage bootloader has not yet initialized your ARM CPU. So, the second-stage bootloader also has to run on the GPU (Graphics Processing Unit).

• The bootloader.bin file is loaded into the 128K 4 way set associative L2 cache of the GPU and then executed. This enables the RAM and loads start.elf which is also in your SD Card.

• This is the third-stage bootloader and is also the most important. It is the firmware for the GPU, meaning it contains the settings or in our case, has instructions to load the settings from config.txt which is also in the SD Card.

• Boot sequence is as follows :

1. At power-up or reset, the ARM CPU is offline.
2. A small RISC core in the GPU begins to execute SoC ROM code (first-stage boot loader).
3. The GPU initializes the SD card hardware.
4. The GPU looks at the first FAT32 partition in the SD media.
5. The second-stage boot-loader firmware named bootcode.bin is loaded into the GPU.
6. The GPU control passes to the loaded bootcode.bin firmware.
7. The file start.elf is loaded by the GPU into RAM from the SD card.
8. An additional file, fixup.dat, is used to configure the SDRAM partition between GPU and ARM CPU.
9. The file config.txt is examined for configuration parameters that need to be processed.
10. Information found in cmdline.txt is presumably also passed to start.elf.
11. The GPU allows the ARM CPU to execute the program start.elf.
12. The module start.elf runs on the ARM CPU, with information about the kernel to be loaded.
13. The kernel is loaded and execution control passes to it.

Q.19 Draw diagram and write algorithm and python code to control LED using Raspberry Pi.

Ans [SPPU : May-19, End Sem, Marks 8]

Ans. : Interfacing of light sensor with Raspberry Pi

- Unlike some other devices the Raspberry Pi does not have any analogue inputs. All 17 of its GPIO pins are digital. They can output high and low levels or read high and low levels.
- For sensors that act as a variable resistor such as LDRs (Light Dependent Resistors) or thermistors (temperature sensors) there is a simple solution. It allows you to measure a number of levels using a single GPIO pin. In the case of a light sensor this allows you to measure different light levels.

• Fig Q.19.1 shows diagram of connecting an LDR to Raspberry Pi.

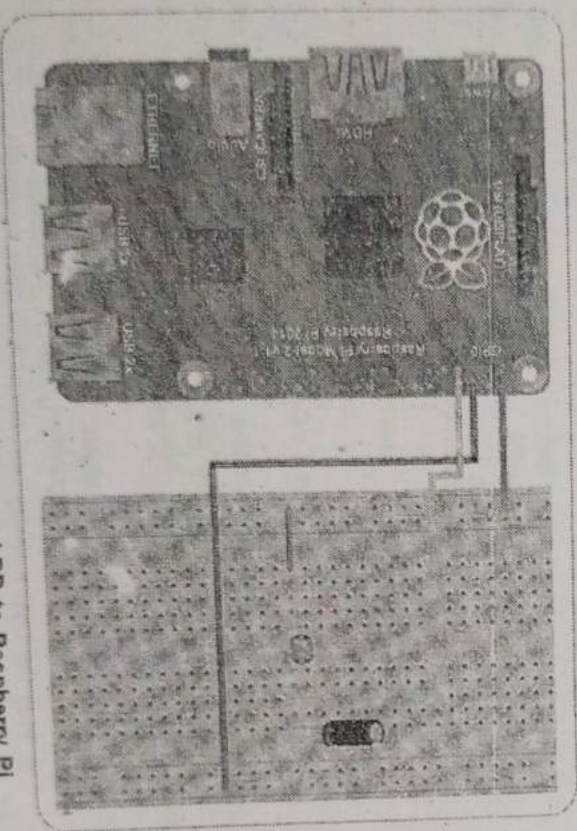


Fig. Q.19.1 : Diagram of connecting an LDR to Raspberry Pi

• Following are steps :

1. First connect pin number 1 (3.3 V) to the positive rail on the breadboard.

2. Next connect pin number 6 (ground) to the ground rail on the breadboard.
3. Now place the LDR sensor onto the board and have a wire go from one end to the positive rail.
4. On the other side of the LDR sensor place a wire leading back to the Raspberry Pi. Hook this to pin number 7.
5. Finally place the capacitor from the wire to the negative rail on the breadboard. Make sure you have the negative pin of the capacitor in the negative rail.

• Fig Q.19.2 shows circuit diagram for above configuration.

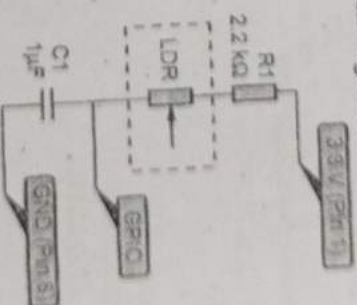


Fig. Q.19.2 : Circuit diagram for LDR

Sequence of events :

1. Set the GPIO pin as an output and set it Low. This discharges any charge in the capacitor and ensures that both sides of the capacitor are 0 V.
2. Set the GPIO pin as an input. This starts a flow of current through the resistors and through the capacitor to ground. The voltage across the capacitor starts to rise. The time it takes is proportional to the resistance of the LDR.
3. Monitor the GPIO pin and read its value. Increment a counter while we wait.
4. At some point the capacitor voltage will increase enough to be considered as a high by the GPIO pin (approx 2 V). The time taken is proportional to the light level seen by the LDR.
5. Set the GPIO pin as an output and repeat the process as required.

Python Code :

```
#!/usr/local/bin/python

# Reading an analogue sensor with a single GPIO pin

import RPi.GPIO as GPIO, time

# Tell the GPIO library to use
# Broadcom GPIO references
GPIO.setmode(GPIO.BCM)

# Define function to measure charge time
def RCTime (PiPin) :
    measurement = 0

    # Discharge capacitor
    GPIO.setup(PiPin, GPIO.OUT)
    GPIO.output(PiPin, GPIO.LOW)
    time.sleep(0.1)

    GPIO.setup(PiPin, GPIO.IN)

    # Count loops until voltage across
    # capacitor reads high on GPIO
    while (GPIO.input(PiPin) == GPIO.LOW) :
        measurement += 1

    return measurement

# Main program loop
while True :
    print RCTime(4) # Measure timing using GPIO4
```

Q.20 Explain Linux on Raspberry Pi.

Ans. : • There are several unix like operating systems for the RPI and there is an operating system called RISC OS that has its origin at the developers of the first ARM chips.

• The Raspberry Pi foundation recommends the use of the following Linux Distributions :

1. Debian 7
2. Raspbian
3. Arch Linux ARM
4. QtonPi

• Raspbian is a free operating system based on Debian optimized for the Raspberry Pi (RPI) hardware.

• The default command prompt on the Pi consists of four components shown in Fig. Q.20.1

```
username      working
              directory
pi@raspberrypi ~ $
hostname      type
              after
              this
```

Fig. Q.20.1 : Command prompt

• Raspbian is the desired operating system for the Raspberry Pi. In order to download and install the operating system onto our Raspberry Pi, you will need Raspbian, Win32Diskmager and USB memory card reader.

1. Download both Raspbian and Win32Diskmager and save somewhere easily accessible.
2. Plug the USB memory card reader into your computer.
3. Open Win32Diskmager.
4. Find the location of the image file and the memory card.
5. Click "Write".

Logging In

PI and at the power lead.

• As soon as we do this, our screen should be black and filled with white text. This will be visible every time we turn on our Raspberry Pi.

• Wait until your screen reads "raspberrypi login."

Username = pi [ENTER]

Password = raspberry [ENTER]

4.5 : Introduction to Arduino Programming

Q.21 What is basic code of Arduino ?

Ans. : Arduino programs run on two basic sections :

```
void setup() {
    //setup motors, sensors etc
}
```

```
void loop() {
    // get information from sensors
    // send commands to motors
}
```

In order to use the Arduino board, we need to connect it to a computer. The setup section is used for assigning input and outputs (Examples : Motors, LED's, sensors etc) to ports on the Arduino. It also specifies whether the device is OUTPUT or INPUT. To do this we use the command "pinMode".

```
void setup() {
    pinMode(9, OUTPUT);
    // 9 is port number
}

2. Loop : The loop contains statements that are executed repeatedly. The section of code inside the curly brackets is repeated depending on the value of variables.
```

Q.22 What is the difference between digitalWrite() and digitalWrite() ?

Ans. : The digitalWrite() function will read the HIGH/LOW value from the digital pin and the digitalWrite() function is used to set the HIGH/LOW value of the digital pin.

Q.23 Write syntax of pinMode().

Ans. : • The specific pin number is set as the INPUT or OUTPUT in the pinMode() function.

• The Syntax : pinMode(pin, mode)

Where,

pin : It is the pin number. We can select the pin number according to the requirements.

Mode : We can set the mode as INPUT or OUTPUT according to the corresponding pin number.

END...